MUHAMMAD ZAIN ABBAS

Full Stack Developer | Next.js, TypeScript, Node.js, PostgreSQL, Python, AI Integration

Islamabad, PK | +92 309 7678854 | iamzainabbass@gmail.com

github.com/ZainAbbas-dev | linkedin.com/in/muhammad-zain-abbas-1a0b87307 | portfolio-gilt-pi-66.vercel.app

PROFILE

Final-year Computer Science student at COMSATS University Islamabad with internship experience building live product features in Next.js and TypeScript. Builds and deploys real-time web applications with Node.js, PostgreSQL, Socket.io, and WebRTC, and serves AI models through Python and FastAPI. Seeking an Associate Software Engineer role.

EXPERIENCE

Hoop Interactive | Web Developer Intern | July 2026 – September 2026

Stack: Next.js, TypeScript, React.js, Node.js, Express.js, REST APIs, Git

- Merged 7 code-reviewed pull requests covering 27 tasks across SEO, backend APIs, and an admin dashboard.
- Built admin queues for verifications, reports, and partners, plus ops analytics and a live audit timeline.
- Implemented SEO structured data, page metadata, and security headers, plus health and waitlist stats APIs.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, C++, Java, Dart
- **Frontend & Mobile:** Next.js, React.js, Tailwind CSS, Flutter
- **Backend & Real-Time:** Node.js, Express.js, FastAPI, Flask, Prisma ORM, REST APIs, Socket.io, WebRTC, JWT
- **Databases:** PostgreSQL, MongoDB, SQLite
- **AI/ML:** PyTorch, scikit-learn, Hugging Face, YOLOv8, OpenCV, DeepFace, TensorFlow Lite, ONNX Runtime
- **DevOps & Tools:** Docker, Git, GitHub, Vercel, Render

PROJECTS

MediQore | Final Year Project: AI-Assisted Offline Health Platform for Lady Health Workers 2026 – 2027

Stack: Flutter, Node.js, Express.js, PostgreSQL, React.js, Leaflet.js, Python, scikit-learn, ONNX Runtime, SQLite

- Co-authored the SRS defining 10 modules and 141 functional requirements for an Android app and web portal.
- Own 5 of 10 modules: Flutter onboarding, registration, visit form, AI risk engine, and the React dashboard.
- Co-designed AES-256 encrypted offline storage and idempotent UUID sync with version-based conflict review.
- Training scikit-learn risk models (SMOTE, SHAP) for on-device ONNX inference, targeting 90% high-risk recall.

CaseRoute | Legal Tech Marketplace | https://case-route.vercel.app/

Stack: Next.js 14, Node.js, Express.js, PostgreSQL, Prisma ORM, Socket.io, Tailwind CSS, JWT

- Built 27 REST API endpoints, 5 Prisma data models, and JWT-secured dashboards for 3 user roles.
- Designed a weighted matching algorithm that ranks lawyers by specialization, experience, and rating.
- Separated Next.js (Vercel) from a stateful Express.js and Socket.io server (Render) for real-time chat.

Nexus Secure Chat | Real-Time Communication Platform | https://nexus-secure-chat-savr.vercel.app/

Stack: React.js, Node.js, Express.js, Socket.io, WebRTC, PostgreSQL, CryptoJS (AES), JWT, Tailwind CSS

- Designed 17 Socket.io events for chat, read receipts, typing indicators, calls, and live collaboration.
- Implemented WebRTC audio/video calls with Socket.io signaling, plus a shared whiteboard and Monaco editor.
- Added AES encryption with per-room keys, regex-based phishing detection, and a 4-tier trust score.

Multimodal Vision Engine | Computer Vision REST API | https://multimodal-vision-engine.vercel.app/

Stack: Python, FastAPI, PyTorch, Hugging Face Transformers, YOLOv8, OpenCV, Docker

- Built 3 FastAPI endpoints serving ViLT (visual QA), Donut (receipt parsing), and YOLOv8 (object detection).
- Deployed with Docker on Hugging Face Spaces; the security scan returns threat alerts with bounding boxes.

EDUCATION

BS Computer Science | COMSATS University Islamabad 2023 – 2027

CGPA: 3.51 / 4.00

CERTIFICATIONS & LANGUAGES

Meta Full Stack Developer (Coursera) | **English** (Professional) | **Urdu** (Native)